

Kadhiravan Gopal

AI/ML & Full-Stack Software Engineer

Chicago, IL • (312) 730-8530 • iitzkadhiravan@gmail.com • linkedin.com/in/kadhiravang • github.com/kadhiravang • kadhiravang.github.io

SUMMARY

AI/ML and full-stack software engineer with **2 years** in LTIMindtree's Generative AI Center of Excellence, working on **8+ full-stack applications** (Python backend, Angular UI) and responsible for them from development through deployment and production support. Hands-on with LLMs (GPT-4o, Claude, Bedrock, Azure OpenAI), RAG, multi-agent systems, vector stores, and evaluation. MS in Artificial Intelligence at Illinois Institute of Technology (graduating Dec 2026); seeking **full-time AI/ML engineering roles**.

WORK EXPERIENCE

Software Engineer, Generative AI Center of Excellence • LTIMindtree

Jul 2022 – Jul 2024

- Trusted to start projects from scratch; responsible for development, debugging, deployment, and live production support across **8+ full-stack applications** (Python backends, Angular UIs).
- Primary developer of an AI-assisted **property insurance underwriting assistant**, built from scratch with a small agile team. It accepted audio, video, PDF, Word, presentation, and image inputs; I wrote the **multi-agent architecture** (manual orchestration, before LangGraph and native function calling existed) that identified each file, processed files in parallel, and produced a property estimate, cutting per-property underwriting from **1–1.5 months of manual work to minutes**.
- Raised RAG accuracy from **70% to 95%** on our test cases by researching and implementing LLM-based chunk re-ranking.
- Stack: Angular, Flask REST APIs, MongoDB, Qdrant, LlamaIndex, LangChain; evaluated GPT-4o, Claude, Mistral, and others. Deployed GPT-4, GPT-4o, and GPT-3.5 Turbo on Azure OpenAI (Key Vault, AI Search, Functions, Document Intelligence) and apps on Azure App Service and AWS.
- On a client project limited to one Azure OpenAI deployment, negotiated a realistic target (8+ simultaneous users, 1–2 minute responses) and built a “canvas” layer that screened malicious or violent requests, tracked requests and tokens per user for usage and pricing, and queued requests within model limits.
- Fixed live production issues (including night incidents); ran knowledge-transfer sessions; presented a handwriting assistant and other apps at IgnAlte.

Student Assistant, CPS Testing Project • Illinois Institute of Technology

Oct 2025 – Present

- Rotate across admin, check-in, and proctoring roles; as admin, collect student and session data, enter it into the testing software, and track it across multi-campus teams.

PROJECTS

HaloCheck • LLM Evaluation & Hallucination Detection (private academic project)

2026

Evaluation harness for LLM-generated clinical summaries using a DeBERTa cross-encoder NLI, scispaCy, medspaCy, and NegEx for entity extraction, negation detection, and severity-weighted factual-fidelity scoring, with regression tracking. Outperformed a published baseline (HHEM): **F1 0.64, Recall 0.83, OOD F1 0.93**.

Agentic RAG Bot • LangGraph, FastAPI, Qdrant, React

2026

Session-isolated RAG over PDFs: a LangGraph agent rewrites, retrieves, grades, and retries; every claim is cited to document, timestamp, and page, and uncovered questions return “not covered” instead of a guess.

FitnessKitchen • AI Nutrition Tracker (Android, Kotlin, Jetpack Compose)

2026

Built end to end: logs meals from text or voice, using Gemini function-calling to ground each entry in USDA nutrition data.

Mini-CLIP and Mini-DiT • Paper Reimplementations in PyTorch (coursework)

2025–2026

Reimplemented CLIP (custom training loop, contrastive loss, zero-shot evaluation, bias/fairness analysis; **~72% top-1** on mini-scale ImageNet) and the Diffusion Transformer paper (ICCV 2023) across MNIST, BDD100K, and CIFAR-10.

TECHNICAL SKILLS

LLMs & Agentic AI: GPT-4o, Claude, Mistral, Amazon Bedrock, Azure OpenAI, Gemini, LangChain, LlamaIndex, LangGraph, RAG, LLM re-ranking, multi-agent orchestration, function calling, prompt engineering, LLM evaluation

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, diffusion models, computer vision, DeBERTa, scispaCy, medspaCy

Engineering: Python (Flask, FastAPI), REST APIs, SQL, Angular, React, Kotlin, Jetpack Compose • PostgreSQL, Oracle, MongoDB, Qdrant, Neo4j (prototype)

Cloud & DevOps: Azure (OpenAI, App Service, Key Vault, AI Search, Functions, Document Intelligence), AWS (Bedrock, ECS, EC2, S3), Docker, CI/CD, Git

EDUCATION

M.S., Artificial Intelligence • Illinois Institute of Technology, Chicago, IL

Jan 2025 – Dec 2026

GPA 3.77/4.0 • Machine Learning, Deep Learning, NLP, Medical Informatics & AI, Computer Vision, Algorithms, Databases

B.E., Computer Science • Panimalar Engineering College (Anna University) • GPA 8.72/10

2018 – Jun 2022

CERTIFICATIONS

Oracle SQL Certified Associate • AWS Partner Accreditations (Technical, Cloud Economics, Sales) • CITI Biomedical Research